Amadeus schema integration: gridMET review

Amadeus Schema Integration at the Storage Boundary

Ordinary Amadeus workflows, explicit provenance recording, and schema enforcement during export and loading.

Preserve science

Reuse existing download, process and calculate routes.

Validate publication

Connect metadata, contracts, provenance, Parquet and DuckDB.

Decide with evidence

Compare correctness, coverage, cost and remaining risks.

Evidence: LATEST, HIST

The proposal keeps Amadeus at the center

Optional integration consumes scientific outputs; it is not a new scientific pipeline.

RESPONSIBILITY	WHO DOES IT	BOUNDARY
Dataset routes and science	Existing Amadeus functions	No namespace patch or new scientific dispatcher
Calls, inputs and recording	Ordinary caller + explicit recorder	Caller still chooses arguments/order; no hidden package hook
Contract and publication	External export/storage support	Can reject publication after calculation
Store inspection	Verified query entry point	Read-only connection after snapshot checks

Existing R integration packages; no runtime Python. Filesystem/I/O changes are explicit.

Evidence: LATEST, CHANGES

Three implementation versions must stay distinct

Ordinary Amadeus is an additional scientific/performance baseline.

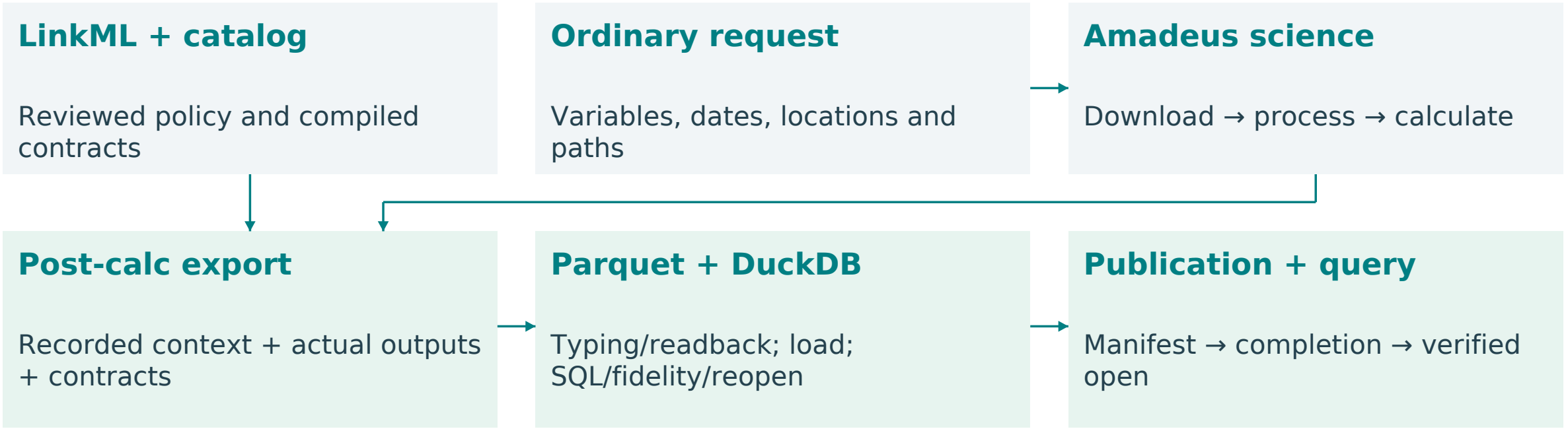
VERSION	WHAT IT ESTABLISHES	WHAT IT DOES NOT ESTABLISH
Optimized pilot	Workflow/generated-R checks; measured in the 45-run study	Latest publication-manifest protections
Measured candidate	Recorded ordinary calls + post-calc storage boundary; same build policy	Consolidated build-status guard or protected saved query
Current corrected release	Guard consolidated; manifest/query integrity; 26 publication tests	A new matched full-workflow runtime/memory comparison

Historical performance labels remain “pilot” and “candidate” for traceability.

Evidence: HIST, LATEST

End-to-end: from request to a verified query

Build definitions once; validate real data when it exists; publish a bound snapshot.



Recording during science; validation after calculate; human approval pending.

Evidence: LATEST, CHANGES

LinkML defines the contract; artifacts carry it to runtime

Source YAML modules are build inputs. Runtime reads compiled contracts and data.

SOURCE MODULE	ROLE IN THE INTEGRATION
amadeus_schema.yaml + common	Composition and shared identifiers/types
catalog	Dataset/product/variable/asset metadata structures
request	Requested variables, locations and time-window structures
provenance + value	Execution/lineage records and reported values
Isolated approval extension	Qualification status, native-unit policy and evidence fields

Generated JSON/R/SQL are pinned. The core schema repository was not edited.

Evidence: CHANGES, LATEST

Metadata has explicit authorities

One released catalog connects source descriptions with actual observed data.

PROPERTY	CURRENT AUTHORITY	LIMIT
Supported codes / labels	Installed Amadeus mapping helpers	33 recognized ≠ 33 onboarded
Native/output units and tolerance	Frozen policy + observed NetCDF units	Scientific review pending; identity only
File identity / grid / source name	Actual source file observations	No claim of continuous immutability
Request and output context	Caller inputs and actual outputs	Not inferred solely from a result table
Coverage, license, external mappings	Only supplied/approved information	Do not invent absent metadata

16 native daily codes are active; PDSI/SPEI/EDDI are not newly qualified here.

Evidence: SCI, LATEST, DIRECT

The user supplies ordinary scientific inputs

The integration adds explicit observation and a final export, not a second dataset route.

Example request

Variables: tmmx, pr
Dates: 10–11 February 2025
Site: Raleigh
Source cache and output paths
No user-authored schema JSON or spreadsheet

Existing Amadeus functions

download_data(dataset_name = "gridmet", ...)
process_covariates(covariate = "gridmet", ...)
calculate_covariates(covariate = "gridmet", ...)

Then: export_gridmet(outputs, request, recorded context, ...)

Scientific profile: point/radius 0, mean, native daily values; no temporal aggregation.

Evidence: LATEST, CURRENTCASES

External R responsibilities are visible and limited

Existing file names are retained; core Amadeus remains independent.

FILE / FUNCTIONS	PURPOSE	WHEN
dev/check_approval.R; validate_release.R	Policy/evidence QA and release fingerprints	Build
provenance_recorder.R	Observe calls/source/output context; verify recording	During science / after calc
export_gridmet.R; schema_export.R	Map supplied results and publish accepted records	After calculate
storage_boundary.R	JSON/semantic/Parquet checks; publication and verification	Write / store open
database.R; parquet_experiment.R	R-native loading, SQL/fidelity/reopen and saved queries	Load / query

No example_gridmet.R or schema_validator.R runtime dependency in the current release.

Evidence: CHANGES, INVENTORY

Validation is placed where its inputs become available

No separate user-facing validation call is required for export or saved queries.

STAGE	WHAT IS CHECKED	WHY HERE
Build	Definitions, code/unit policy, evidence, compiled artifact agreement	Stable facts before distribution
Scientific execution	Amadeus argument checks; explicit recording	Observe history; no schema gate injected into science
After calculation / write	Actual outputs, source units, membership, schema, lineage, Parquet readback	Actual data now exists
Load / saved open	SQL/fidelity/reopen; then completion + byte identity	Validate storage and detect later changes

Parquet typing supports the contract; semantic rules and actual-data checks still execute.

Evidence: LATEST, CHANGES

One value has a traceable record chain

Example: Raleigh maximum temperature, 10 February 2025 = 279.1 K.

User request

tmmx / Raleigh / date window

→

Calculation run

Function, arguments, role, time, output

→

Value + asset

279.1 K; request/variable/run links

Source identity

tmmx_2025.nc; SHA b8583ca388c0...

→

Recording evidence

Execution RDS + events + seal

→

Published snapshot

Instance + Parquet + DB + receipts

Observed context is stronger than a plausible record; the local chain is still unsigned.

Evidence: LATEST, AUDIT, DIRECT

Parquet preserves typed records and supports efficient access

Here, Parquet I/O uses DuckDB; Arrow/nanoparquet are not compared.

What it contributes

Typed extracted-value rows and compact storage. Source grids remain assets. Readback compares written records with accepted values.

What still needs rules

Units, references, request membership, lineage, missingness and scientific meaning are not proven by typing.

How it is connected

Pinned contracts, catalog/instance identities and receipts bind the file to the release and database.

Boundary and adapter both materialize Parquet; no zero-copy or free-validation claim.

Evidence: LATEST, CHANGES

Publication and query verification are separate phases

A completed database publication is more than schema acceptance.

1. Load and verify

DDL → 102 SQL → fidelity → reopen

2. Bind snapshot

Manifest hashes owned data + receipts

3. Complete

Read-only files → workflow → LATEST

4. Verify at store open

Successful release + completion + manifest + file/receipt bindings

5. Query read-only

No provider NetCDF reread; edited snapshot rejected

Failed load: schema outputs may remain; no completed workflow or pointer advance.

Evidence: LATEST, PUBTEST, FAIL

Pilot versus current storage-boundary integration

Both preserve the tested scientific route; their enforcement architecture differs.

AREA	OPTIMIZED PILOT	CURRENT STORAGE-BOUNDARY RELEASE
Scientific calls	Existing Amadeus routes, caller-coordinated	Same routes; ordinary caller + recording
Validation	Workflow/raster handoffs + generated R/JSON checks	After calculate: JSON/shared R/SQL + write/readback
Metadata / provenance	Original mapping and call records	Package labels/catalog + verified recorded context
Saved database	Historical saved-query trust gap	Manifest/completion checks and byte verification
Maintenance scope	Workflow check orchestration	Publication protocol + explicit metadata/profile

Neither implementation changes core Amadeus. Dataset-specific policies remain explicit.

Evidence: HIST, LATEST, CHANGES

Matched validation tests show bounded agreement

Evidence from the frozen pilot/candidate comparison; newer integrity tests are separate.

TEST	OBSERVED RESULT	INTERPRETATION
251 invalid mutations per arm	502 rejections + 2 valid controls = 504 outcomes	Structural/semantic fixture agreement
Focused historical gaps	7 invalid cases rejected per arm	Geometry, interval, clock label, binding, membership, producer, cycle
Candidate recording checks	16 checks passed	Supplied context/output/evidence tampering and lifecycle
Model coverage	11 of 17 collections populated	102 SQL checks are not full-model coverage

Arbitrary descriptive text remains accepted; validators do not establish scientific truth.

Evidence: PARITY, SEMANTIC, RECORD, COVER

Latest corrected release: end-to-end evidence

Fresh scientific executions used identical cached sources, not new provider downloads.

EVIDENCE	RESULT	WHAT IT SUPPORTS
Fresh execution cases	4, 336 and 4 accepted results; 102 SQL each	Complete typed/storage/query path
Native regression	Maximum difference from ordinary reference = 0	Preserved tested values
Publication checks	26 passed	Changed DB/Parquet/receipts, status, coverage, relocation
Independent ncdf4 reads	336 values within declared tolerances	Source mapping and packed-value decoding
Load failure and retry	Incomplete publication rejected; fresh retry passed	Bounded recovery; not power-loss proof

Source-copy removal and schema-only export were also tested; original cache untouched.

Evidence: LATEST, CURRENTCASES, PUBTEST, DIRECT, AUDIT, FAIL

Performance study: controlled comparisons, explicit limits

45 completed processes: 3 arms × 3 request sizes × 5 repetitions.

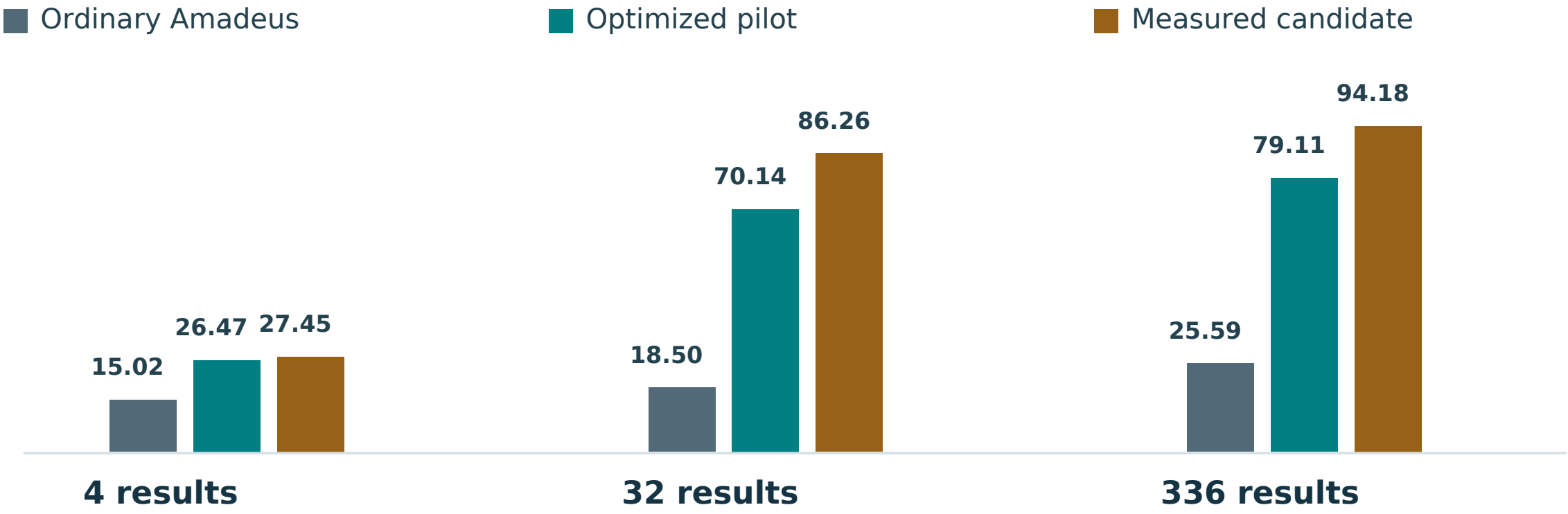
DESIGN	DETAIL
Request sizes	4 values; 32 values; 336 values
Scientific controls	Same calls/arguments, source bytes and TIFF handoff
Execution controls	Fresh R processes; randomized arm order within matched blocks
Measurements	Whole-process elapsed; main HWM; sampled process-tree RSS; output bytes
Limits	Warm filesystem/cached sources; shared host; request-cardinality scaling

Interrupted attempt excluded; completed runs retained; N4 sensitivity is reported.

Evidence: HIST, PERF, SENS

Historical runtime: integration has a measurable cost

Whole-process medians in seconds; corrected publication release is not timed here.



Added cost buys metadata, provenance, validation and storage; it is not isolated validation cost.

Evidence: PERF, PAIRED

Historical memory: report different measures separately

Largest request: 336 results; five executions per arm.

MEASURED ARM	MEDIAN MAIN HWM, MiB	MEDIAN TREE RSS PEAK, MiB
Ordinary Amadeus	669.4	665.0
Optimized pilot	732.2	968.2
Measured candidate	1039.5	1035.1

Main high-water memory and sampled process-tree RSS are not interchangeable. Tree sums can double-count shared pages and miss peaks.

Evidence: PERF

Current integrity checking adds a scoped store-open cost

Warm-cache component measurements; not full query startup or workflow speed.

About 0.13 seconds

Median verification time for each tested snapshot size; three samples per size.

About 15–16 MB

11 owned artifacts verified. Original annual NetCDF files are not re-read.

Once per open

Each CLI invocation opens/verifies. No trusted stat-only cache or per-statement rehash.

Large-store cost, cold-cache behavior and production concurrency still need qualification.

Evidence: COST, LATEST, SOURCELESS, SCHEMAONLY

Remaining gaps are decision inputs

Technical evidence is strong within scope; it is not production certification.

AREA	CURRENT LIMIT	REQUIRED ACTION
Scientific meaning	Units/physical intervals/method suitability pending	Human review of scoped policy and evidence
Portability	Successful host; colleague/affected stack not replayed	Independent request and environment capture
Trust and concurrency	Unsigned chain; read-only bits do not stop privilege	Choose actual immutable/access-control policy
Durability and migration	Bounded retries; no power-loss proof; old stores uncertified	Agree durability/migration requirements
Coverage and cost	16 codes; point profile; DuckDB; latest total performance unmeasured	Qualify broader uses and acceptable budgets

Current production flag remains false; no reviewer success has been prefilled.

Evidence: CURRENTGAPS, SCI, REPLAY, STEPS

External metadata can be added without runtime spreadsheets

Future build-time modes; no external import/merge is implemented in this release.

MODE	BUILD-TIME APPROACH	NON-NEGOTIABLE REQUIREMENTS
Amadeus-derived baseline	Package codes/labels + frozen policy + observed source facts	Do not invent absent coverage/units
External catalog baseline	Reviewed catalog properties normalized to the same target	Still confirm supported Amadeus routes; preserve observations
Combined catalog	Keyed merge with explicit field authority and conflict decisions	No silent last-source-wins policy; one released catalog

Stable keys, versions/hashes, required fields, unit meaning, scope evidence and review are required.

Evidence: LATEST, SCI

Decisions before wider adoption

Adopt the architecture only with an explicit scope, trust model and measured cost.

DECISION	EVIDENCE / OPEN GATE
Accept the external post-calc boundary?	Core unchanged; published-output enforcement demonstrated
Approve the native scientific profile?	336 source comparisons support review; human units/time/scope pending
Which environment is supported?	Capture identities and replay on independent/affected stack
What protects published snapshots?	Local integrity works; choose deployment immutability/durability policy
What goes to the review branch?	Current code, tests, docs, scoped evidence; exclude raw cache and tamper copies

Review → independent replay/sign-off → limited shadow use → production decision.

Evidence: STEPS, LATEST, CURRENTGAPS

A reproducible demonstration for the team

The review pack includes a portable snapshot and matching integration code.

OPEN / RUN	WHAT TO SHOW
demo-files/snapshot/results.csv	Four native results: request keys, dates and values
schema-instance.json + source-metadata.json	Catalog/variable/units, source hash and record links
publication-manifest.json + workflow receipt	Completion and file/release bindings
DEMO-WALKTHROUGH.txt	Supported verified query of the bundled binary DB
review worksheets + TEAM-REPLAY guide	Pending scientific approval and independent replay

Copied code and relocated snapshot were queried successfully; no provider download.

Evidence: LATEST, PUBTEST

Appendix: scientific and metadata limits

Use the complete worksheets and gap register for decisions.

AREA	LIMIT AND REVIEW IMPLICATION
Scientific approval	Numeric source agreement is not provider truth or scientific meaning approval.
Physical time	CF date coordinate/label agreement does not establish daily physical intervals.
Scope	16 native daily codes; point/radius 0 / mean; not all33 codes or methods.
Model coverage	Unpopulated classes and planned operational schemas/migrations remain unqualified.
Metadata authority	Package mapping is not complete catalog authority; external import deferred.
Missingness / source lifecycle	Finite policy is scoped; saved queries do not prove continued raw-file availability.

Evidence: CURRENTGAPS, SCI, DIRECT

Appendix: operational and performance limits

Local byte integrity does not establish all deployment guarantees.

AREA	LIMIT AND REVIEW IMPLICATION
Unsigned trust / concurrency	Privileged coordinated rewrites and concurrent path replacement are not covered.
Durability / recovery	Pointer rename and bounded retry tests do not prove multi-file power-loss durability.
Historical compatibility	Old stores need matching release or explicit requalification; no silent certification.
Access path	Direct DBI/IDE or unrelated external-file SQL bypasses snapshot guarantees.
Filesystem / environment	chmod/rename capabilities explicit; ACL/NFS/Windows and independent stack pending.
Performance / backend	No latest N5 full-workflow benchmark; no Sedona, large-store or multi-source join qualification.

Evidence: CURRENTGAPS, COST, ENV

Appendix: evidence and review package index

Source copies, build source, editable diagrams and presenter notes travel with the deck.

LOCATION	CONTENTS
evidence/ + evidence-sources.json	Copied CSV/JSON evidence; original paths and SHA256
implementation/code/integration/	Current runtime/build code and pinned contracts
implementation/*review.zip	Complete corrected release and scientific/replay materials
workflow-complete.mmd / .svg / .html	Editable Mermaid, vector diagram and offline visual
presenter-notes.txt; build_review.py	Explanation/source pointers; reproducible authoring source

Slides are editable shapes/text; PDF and previews are static. No repository edits or push.

Evidence: LATEST, HIST

AI assists development; humans approve meaning

Every rule needs an authority, implementation, test evidence and human decision.

AI ASSISTANCE	DETERMINISTIC EXECUTION	HUMAN RESPONSIBILITY
Draft mappings, checks and tests	LinkML build and pinned contracts	Approve metadata and scientific definitions
Review code and explain evidence	R / JSON Schema / SQL after calculate	Approve units, physical time and binding
Surface gaps and summarize results	Manifest verification at saved-store open	Review provenance truth and acceptance policy
Document changes and limitations	No live AI model judges results	Decide release scope; approvals remain pending

Test passes, build success and AI review do not constitute human scientific approval.

Trace a rule from authority to a human decision

Use the supplied registers; preserve unknowns instead of inventing approval.

RECORD	WHAT TO KEEP
Authority	Source/version/hash; observed fact versus reviewed scientific policy
Implementation	Stable rule ID and actual R / SQL / compiled contract location
Test evidence	Control, invalid case, outcome, version and coverage limits
Contribution and approval	Known AI assistance; human edits; reviewer/date/decision/scope

All storage SQL rules enumerated; individual negative-test linkage still needs review.

Follow each stage without adding runtime AI

Documentation-only update: scientific calls, runtime code and evidence are unchanged.

PHASE	EXECUTION / DEPENDENCIES	REVIEW POINT
Build	LinkML and build tools; compile contracts	Scientific policy, metadata authority and scope
Science + recording	Existing Amadeus / spatial stack; explicit recorder	Actual inputs, source identities and history
After calculation	R / JSON Schema / SQL; typed Parquet; DuckDB	Binding, acceptance and readback evidence
Saved open + release	Hashes / DBI / DuckDB; separate human review	Integrity, independent replay and release decision

See stage-traceability.csv and dependency-and-execution-register.csv.